

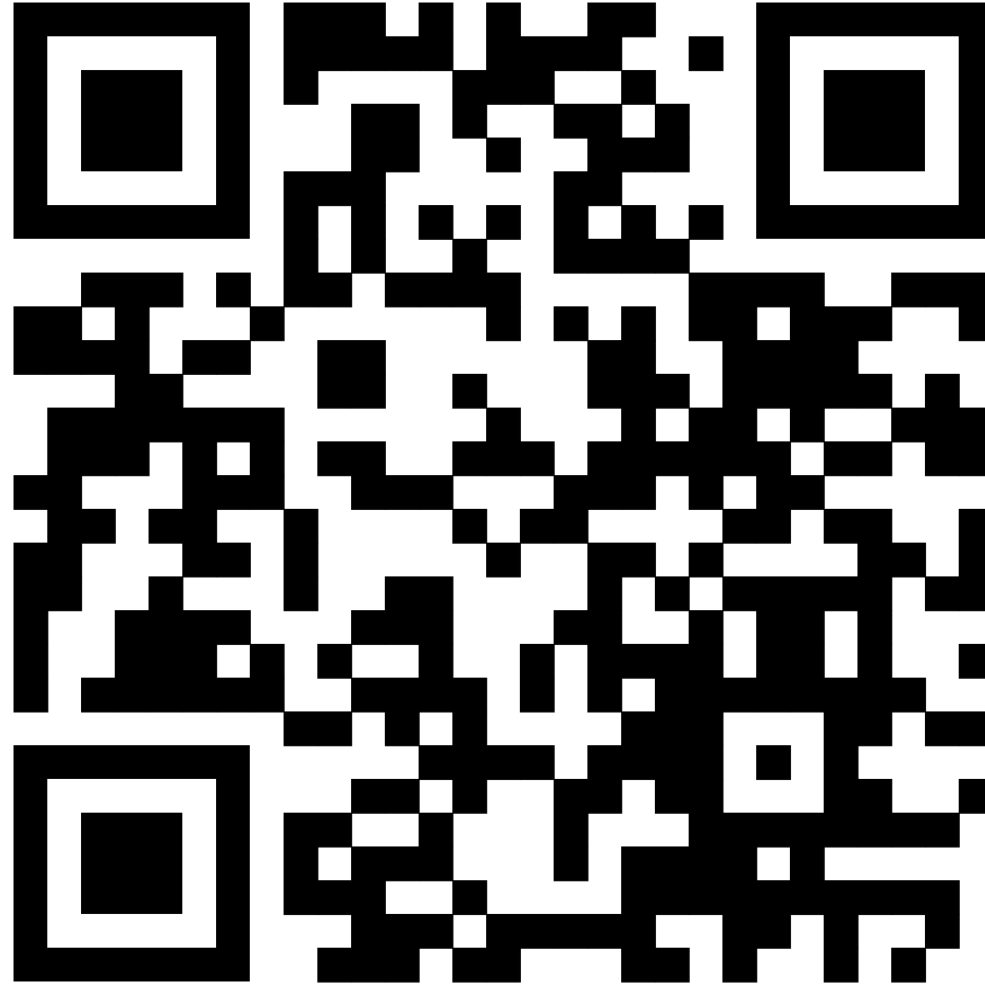
ROB 5014 Math and Simulation for Robotics Introduction

Hyungpil Moon, Professor



SUNGKYUNKWAN
UNIVERSITY (SKKU)

Department of Mechanical Engineering
Department of Intelligent Robotics

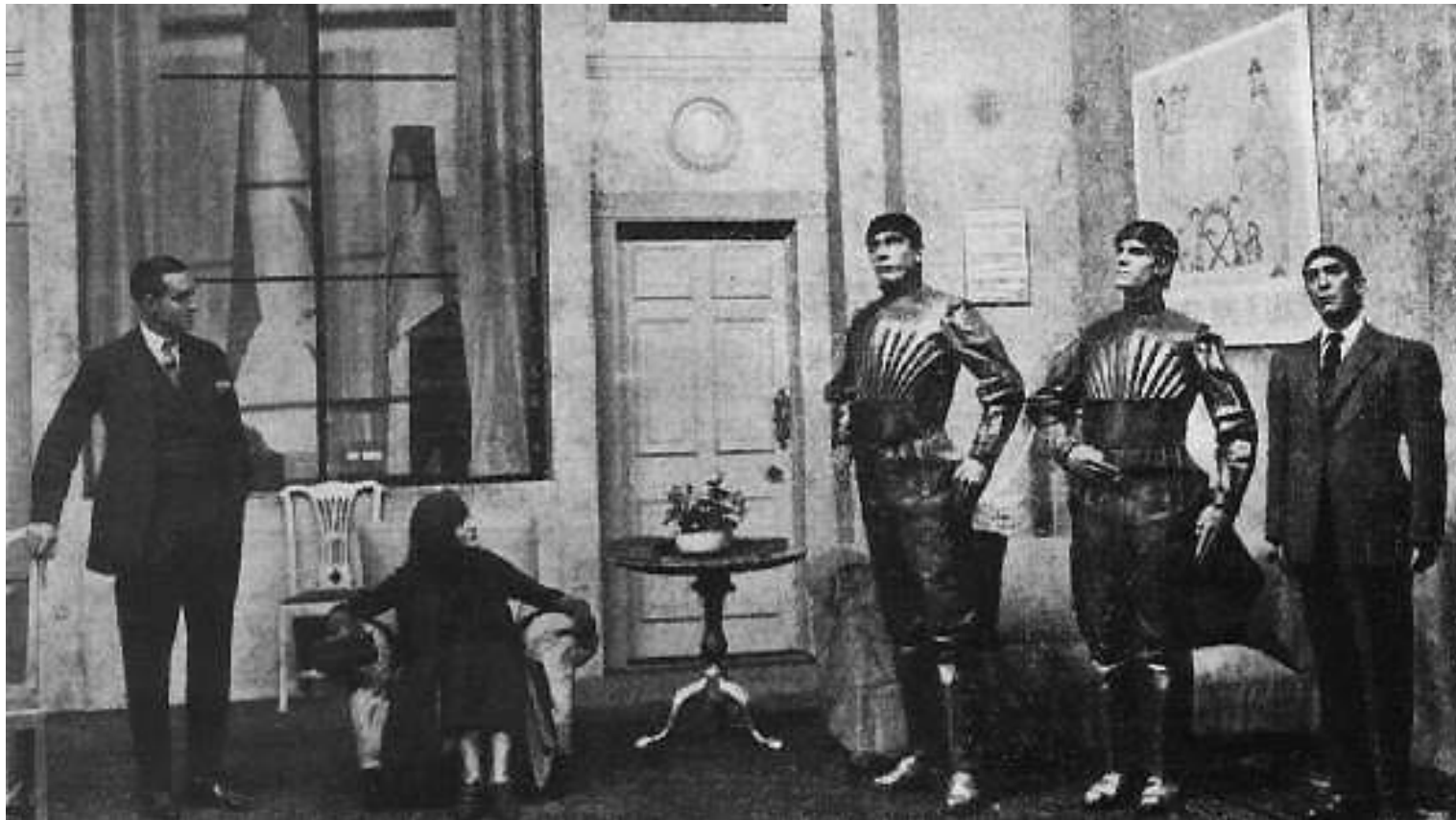


“AMAZING THINGS ARE HAPPENING IN ROBOTS”

Humanoid, Industry 4.0, Drones, Home service,
Autonomous Vehicles, Logistics, ...

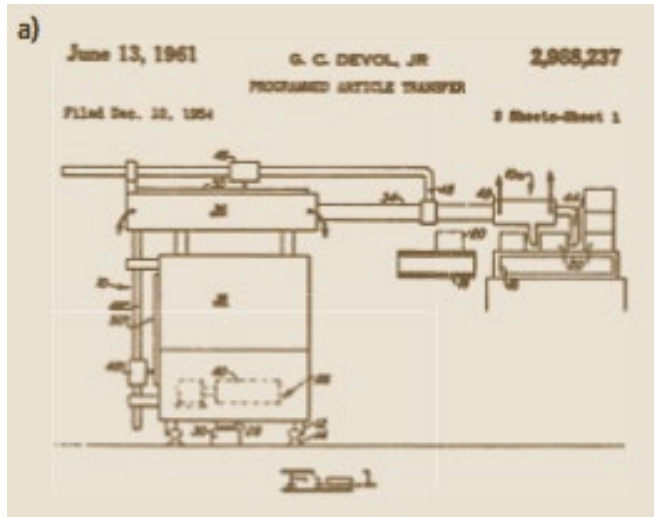
- “**Machines** for **Intelligent** connection between **perception** and **action**”
 - RIA = Robotic Institute of America
 - **Re-programmable** multi-functional **manipulator** designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks, which also acquire information from the environment and move intelligently in response
- ISO 8373
- Automatically controlled, reprogrammable, multipurpose **manipulator** programmable in three or more axes, which may be either fixed in place or mobile for use in industrial automation applications

- “Robot” Slavic word for worker(1921)
 - A Czecho play, titled “Rossum’s Universal Robots(RUR)” written by Karel Capek
 - Not a machine, but a biologically generated human-like creature



- Unimate

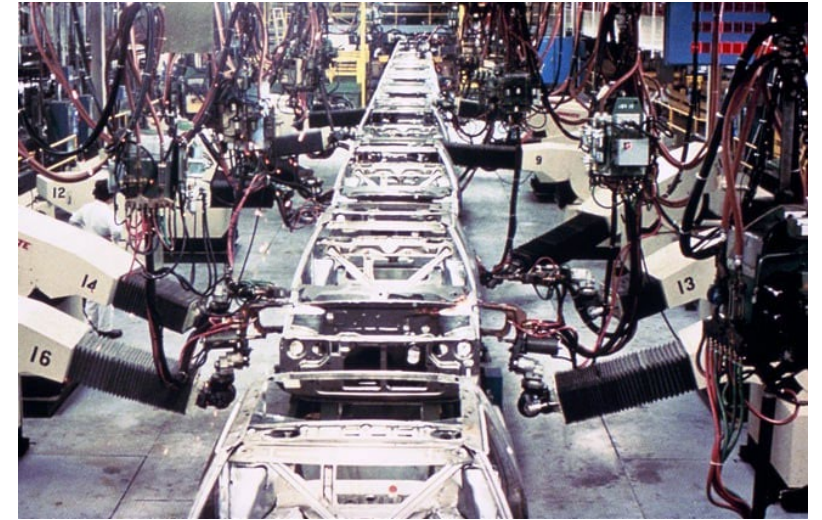
- Invented in 1954
- Installed in General Motors in 1961
- Die casting, welding



(US Patent, G. Devol and J. Engelberger, Unimation)



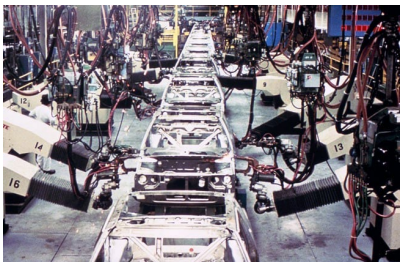
(The Original Unimate)



1969 GM Spot-welding robot

- PUMA

- The **P**rogrammable **U**niversal **M**anipulation **A**rm, called **PUMA**, was introduced on 1975.
- Unimation → Westinghouse → Staubli
→ Kawakaki robotics
- Paint, welding, glass handling
- Semi-conductor wafer handling
- Collaborative Robots



1969 GM Spot-welding robot



1978 PUMA 6DOF



1984 SCARA robot



1992 Delta robot



1999 Da Vinci

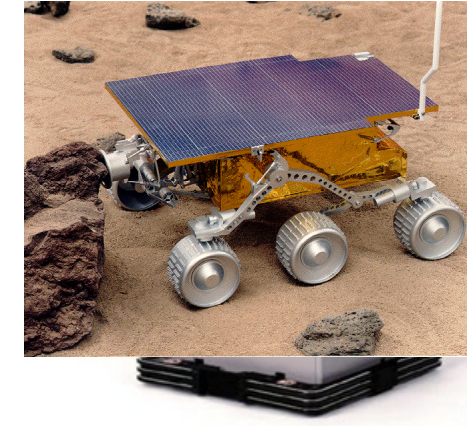


2006 KUKA LWR



2008 Universal Robots

- Lab
 - 1966: Shakey the robot (SRI)
- Hospital
 - 1988: The first HelpMate service robot went to work at Danbury Hospital in Connecticut.
- Mars
 - 1997: NASA's PathFinder landed on Mars. The wheeled robotic rover sent images and data about Mars back to Earth.
- Combat field
 - 2001: iRobot Packbots searched through the rubble of the World Trade Center. Subsequent versions of the Packbot robots are used in Afghanistan and Iraq.
- SLAM paper TRO 2001: A solution to the simultaneous localization and map building (SLAM) problem
- House
 - 2002: iRobot released the first generation of Roomba robotic vacuum cleaners.
- DARPA Challenges
 - 2004, 2005: DARPA Grand Challenge
 - 2007: DARPA Urban Challenge
- Finally, technologies are coming to streets
 - 2014: Tesla Autopilot: <https://youtu.be/C3DbrYx-SN4>
 - 2017: Autopilot 2.0 Local roads

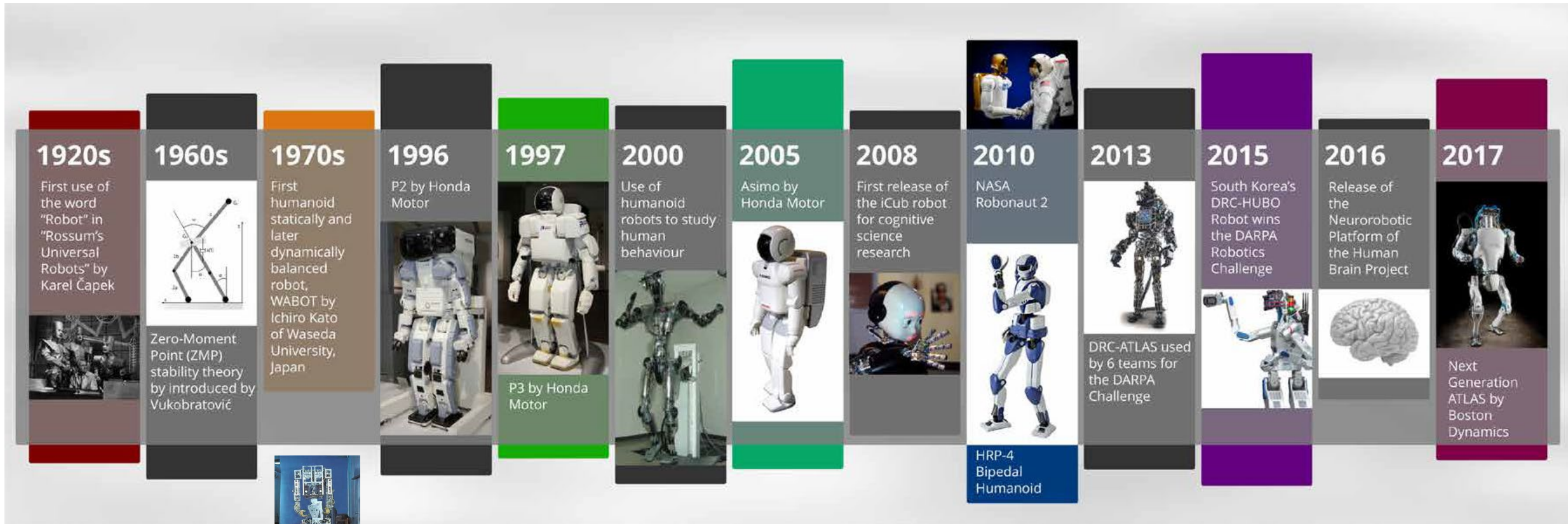


Humanoid Robots



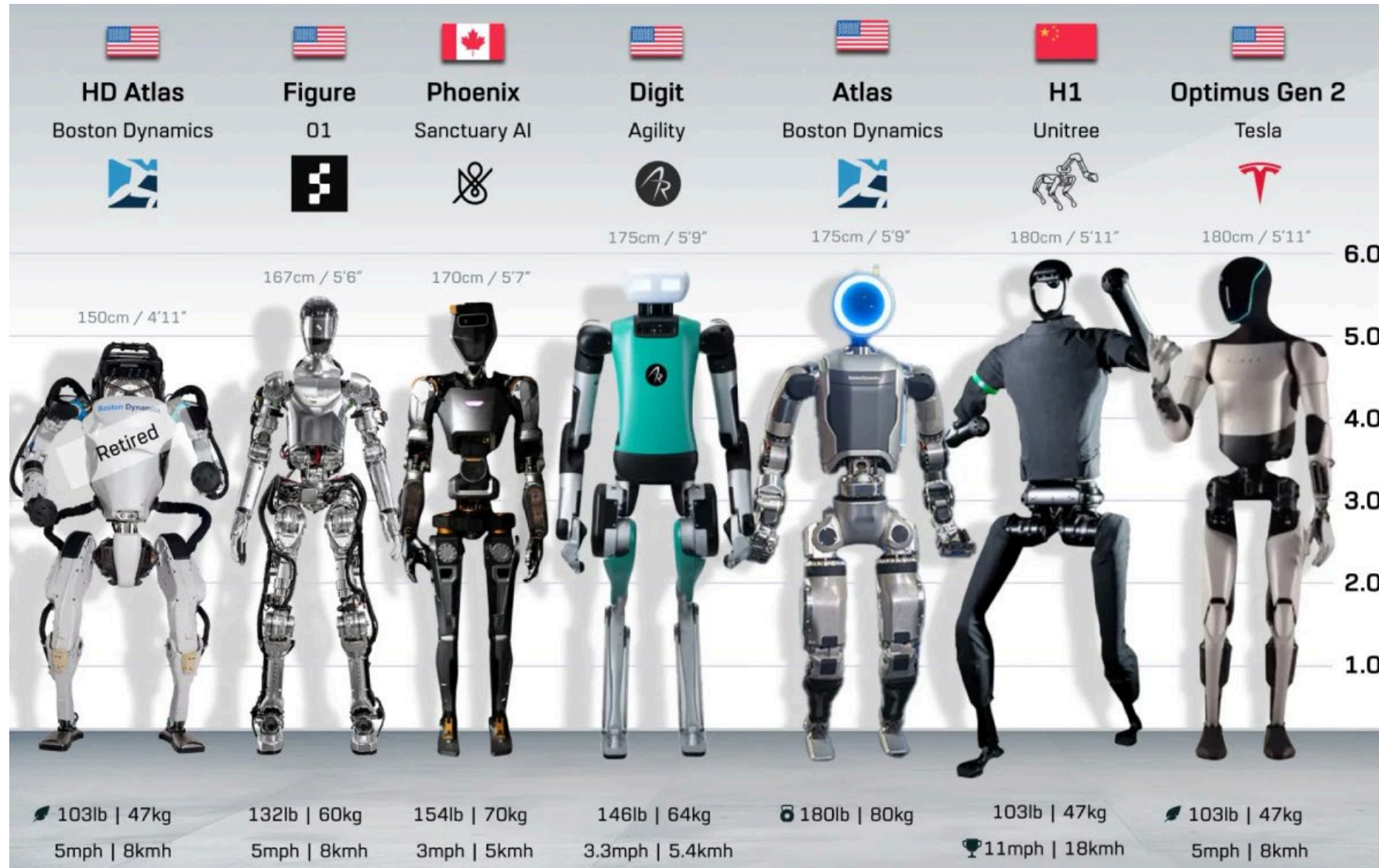
RISE LAB
Robotics and Intelligent System Engineering

SUNG KYUN KWAN
UNIVERSITY (SKKU)



T. Fukuda, P. Dario, G.-Z. Yang, Humanoid robotics—History, current state of the art, and challenges. *Sci. Robot.* 2, eaar4043 (2017).

Humanoid Robots 2024

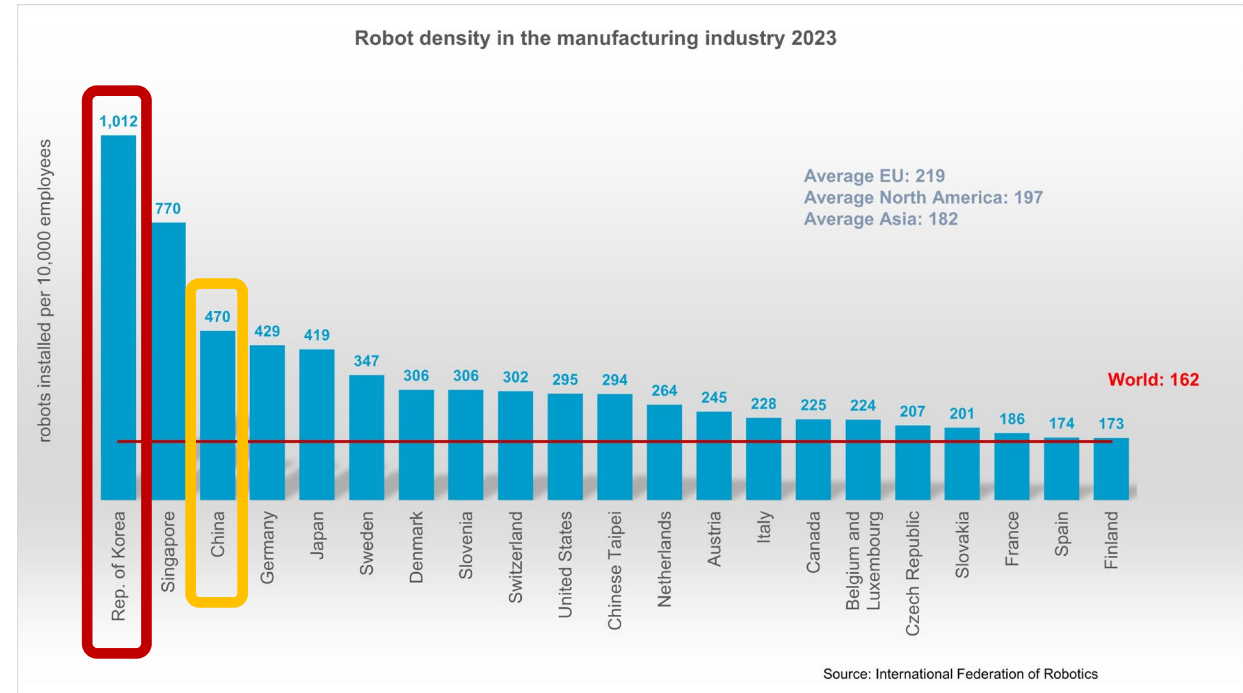
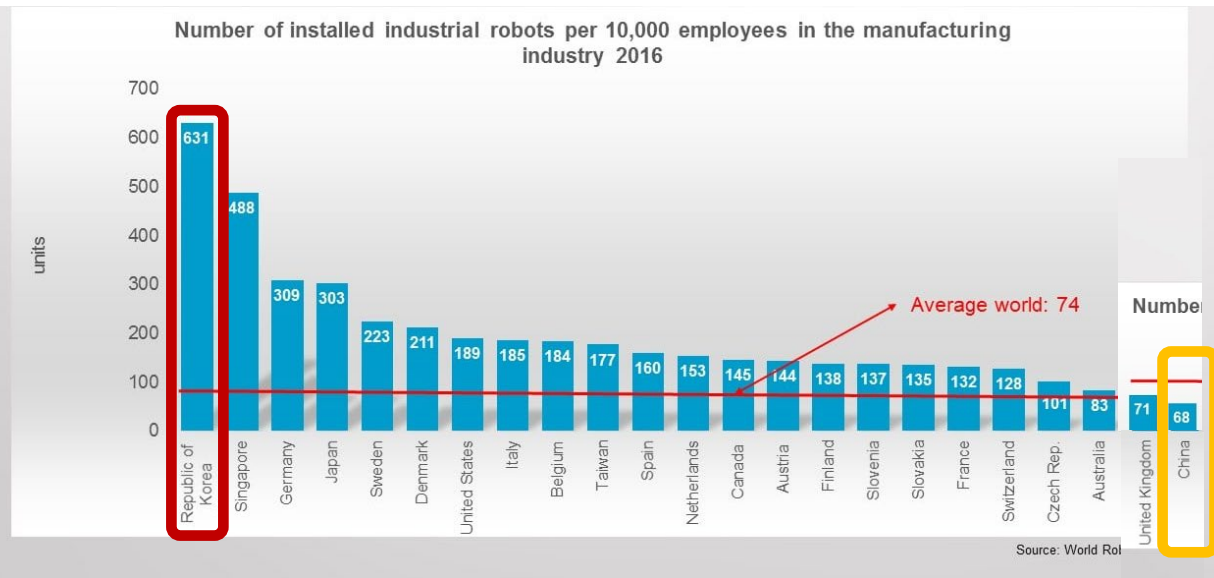


<https://www.visualcapitalist.com/cp/visualizing-the-current-generation-of-humanoid-robots/>

IFR Robot Density



RISE LAB
Robotics and Intelligent System Engineering



Success and Failure



RISE LAB
Robotics and Intelligent System Engineering

SUNG KYUN KWAN
UNIVERSITY(SKKU)

Failure

Success

2017 Mask RCNN by Facebook
2020 Vi-transformer by Google
2025 DeepSeek-R1 by DeepSeek



ROBEAR



아이보

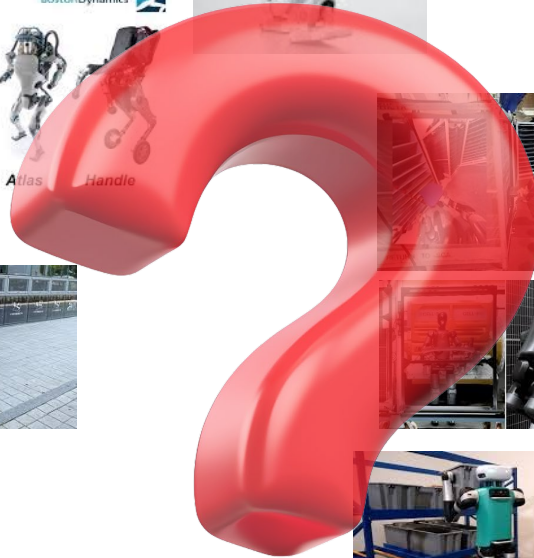


Paro



Lovot

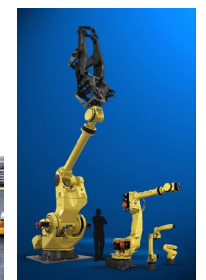
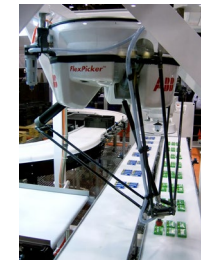
Platforms



Digit



WIM



Syllabus

ROB5014-41 Mathematics and Simulation for Robotics (로봇수학및시뮬레이션)

Sungkyunkwan University

Department of Intelligent Robotics

Spring 2025

(Revision date: March 3, 2025)

General Information:

Lectures: 혁신수업

41: Room 26110 Tuesday 10:30~11:45, and Thursday 09:00~10:15,

Instructor: Prof. Hyungpil Moon

Office: #21321

Office Hours: Thursday 10:30~11:30 or with appointment

Phone: 031-299-4842, Email: hyungpil@skku.edu

TA: TBA

TA Office Hours: Friday 14:00~15:00 or with appointment (Office #24205, Tel: 031-299-4685)

Exam schedule:

Mid-term exam: 09:00~10:15, Apr. 17(Thu) Icampus

Final exam: TBD

Course Contents:

This course deals with mathematical fundamentals for intelligent robotics such as linear algebra, probability, statistics, optimization for robotics. The mathematics concept is implemented in computer programming. ROS, robotic simulators such as Isaac Sim is covered. Robotic kinematics is practiced with robot simulation

Textbook: Course notes: 로봇수학 및 시뮬레이션 ISBN: 979-11-93673-06-5 (95550)

References:

A. Aldo Faisal, Cheng Soon Ong, Mathematics for Machine Learning, Cambridge University Press, 2020. (<https://mml-book.github.io/book/mml-book.pdf>)
F.C. Park and K.M. Lynch, Introduction to Robotics: Mechanics, Planning, and Control,

Computer:

Each student is required to use his/her own notebook computer where Linux, ROS2, Isaac Sim/Lab and more can be installed. No computers will be provided from the class. It is noted that the performance of the computer may affect the quality of the project work.

[Minimum spec]

CPU: Intel i5-11400 or AMD Ryzen 5 5600

GPU: Nvidia RTX 2070 (Note: AMD graphics cards and Nvidia GTX series are not compatible.)

RAM: 16 GB

OS: Linux Ubuntu 22.04

Storage: 50 GB for Isaac Sim

[Recommended spec]

CPU: Intel i7 or AMD Ryzen 7

GPU: Nvidia RTX 4060

RAM: 32 GB

Grading:

Coding reports/HW (70%)

Midterm exam (20%)

Final project (10%)

Exam and Report Policy:

Unless arrangements have been made ahead of time and approved by the instructor, there will be NO make-up exam, NOR will late submission of the report be accepted.

All the reports are submitted to I-CAMPUS system.

Breaking honor code will be treated seriously.

The policy of each exam is CLOSED-BOOK. The allowed item in each exam is a calculator. No smartphone is allowed.

ROB5014-41 Mathematics and Simulation for Robotics (로봇수학및시뮬레이션)
Schedule (subject to change)

Week	Date	Lecture (AV or lecture)	Activity	Note
1	03/04 03/06	Survey, Introduction Numerical integration	Math coding/HW1	
2	03/11 03/13	Forward Kinematics	Math coding/HW2	
3	03/18 03/20	Camera calibration	Math coding/HW3	
4	03/25 03/27	Geometric calibration	Math coding/HW4	
5	04/01 04/03	Hand-eye calibration	Math coding/HW5	
6	04/08 04/10	Graph SLAM	Math coding/HW6	
7	04/15 04/17	Inverse Kinematics with inverse/transpose	Math coding/HW7	
8	04/22 04/24	Regularization	Math coding/HW8	동제로춘계학회
9	04/29 05/01	Linear programming	Math coding/HW9	
10	05/06 05/08	Midterm exam		어린이날
11	05/13 05/15	Introduction to Isaac Sim		국과위
12	05/20 05/22	Isaac Sim ROS Bridge	Physics Solver/Franka joint control (Mobile robot navigation)	ICRA
13	05/27 05/29	Isaac Sim Nav2	Mobile robot navigation Numerical integrator lecture	
14	06/03 06/05	RL in a Nutshell Introduction to Isaac Lab	Cart-pendulum control	
15	06/10 06/12	Issac Lab RL	PID-PPO comparison	
16	06/17 06/19	Final Project Reporting		